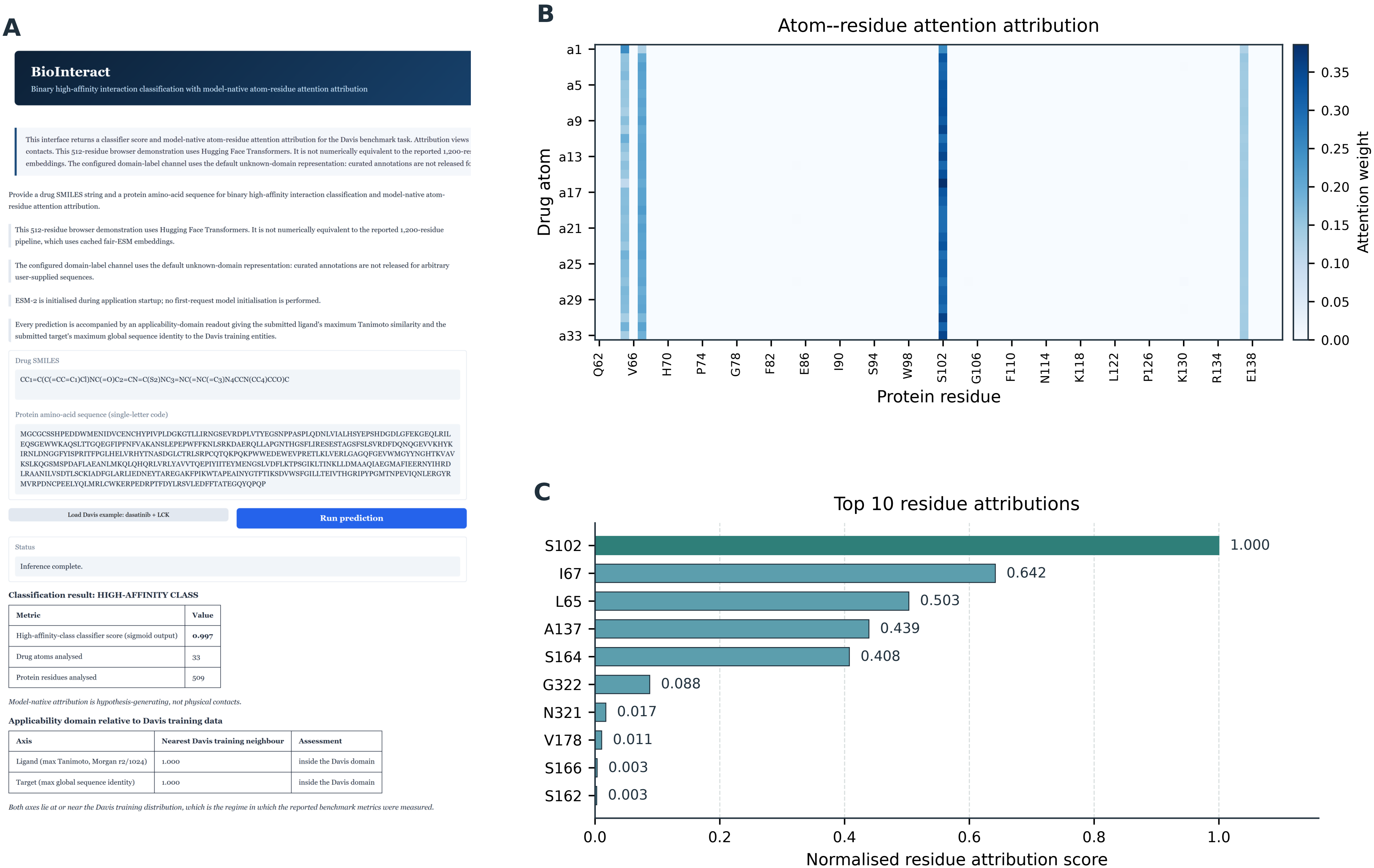
A
BioInteract
Binary high-affinity interaction classification with model-native atom-residue attention attribution
This interface returns a classifier score and model-native atom-residue attention attribution for the Davis benchmark task. Attribution views
contacts. This 512-residue browser demonstration uses Hugging Face Transformers. It is not numerically equivalent to the reported 1,200-re
embeddings. The configured domain-label channel uses the default unknown-domain representation: curated annotations are not released f
Provide a drug SMILES string and a protein amino-acid sequence for binary high-affinity interaction classification and model-native atom-residue attention attribution.
This 512-residue browser demonstration uses Hugging Face Transformers. It is not numerically equivalent to the reported 1,200-residue pipeline, which uses cached fair-ESM embeddings.
The configured domain-label channel uses the default unknown-domain representation: curated annotations are not released for arbitrary user-supplied sequences.
ESM-2 is initialised during application startup; no first-request model initialisation is performed.
Every prediction is accompanied by an applicability-domain readout giving the submitted ligand's maximum Tanimoto similarity and the submitted target's maximum global sequence identity to the Davis training entities.
Drug SMILES
CC1=C(C(=CC=C1)Cl)NC(=O)C2=CN=C(S2)NC3=NC(=NC(=C3)N4CCN(CC4)CCO)C
Protein amino-acid sequence (single-letter code)
MGCGCSSHPEDDWMENIDVCENCHYPIVPLDGKGTLLIRNGSEVRDPLVTYEGSNPPASPLQDNLVIALHSYEPSHDGDLGFEKGEQLRIL
EQSGEWWKAQSLTTGQEGFIPFNFVAKANSLEPEPWFFKNLSRKDAERQLLAPGNTHGSFLIRESESTAGSFSLSVRDFDQNQGEVVKHYK
IRNLDNGGFYISPRITFPGLHELVRHYTNASDGLCTRLSRPCQTQKPQKPWWEDEWEVPRETLKLVERLGAGQFGEVWMGYYNGHTKVAV
KSLKQGSMSPDAFLAEANLMKQLQHQRLVRLYAVVTQEPIYIITEYMENGSLVDFLKTPSGIKLTINKLLDMAAQIAEGMAFIEERNYIHRD
LRAANILVSDTLSCKIADFGLARLIEDNEYTAREGAKFPIKWTAPEAINYGTFTIKSDVWSFGILLTEIVTHGRIPYPGMTNPEVIQNLERGYR
MVRPDNCPEELYQLMRLCWKERPEDRPTFDYLRSVLEDFFTATEGQYQPQP
Load Davis example: dasatinib + LCK
Run prediction
Status
Inference complete.
Classification result: HIGH-AFFINITY CLASS
Metric | Value
High-affinity-class classifier score (sigmoid output) | 0.997
Drug atoms analysed | 33
Protein residues analysed | 509
Model-native attribution is hypothesis-generating, not physical contacts.
Applicability domain relative to Davis training data
Axis | Nearest Davis training neighbour | Assessment
Ligand (max Tanimoto, Morgan r2/1024) | 1.000 | inside the Davis domain
Target (max global sequence identity) | 1.000 | inside the Davis domain
Both axes lie at or near the Davis training distribution, which is the regime in which the reported benchmark metrics were measured.
B
Atom--residue attention attribution
Drug atom
a1
a5
a9
a13
a17
a21
a25
a29
a33
Protein residue
Q62
V66
H70
P74
G78
F82
E86
I90
S94
W98
S102
G106
F110
N114
K118
L122
P126
K130
R134
E138
Attention weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
C
Top 10 residue attributions
S102 1.000
I67 0.642
L65 0.503
A137 0.439
S164 0.408
G322 0.088
N321 0.017
V178 0.011
S166 0.003
S162 0.003
0.0
0.2
0.4
0.6
0.8
1.0
Normalised residue attribution score